



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

End-to-End Supply Chain and Operational Performance Analysis Across Petroleum, Chemical, and Transportation Sectors Using Data Analytics

The domain of the Project:

COURSE NAME

Data Analytics

Team Mentors (and their designation):

V Keerthana

Team Members:

Ms. Brungi Sreeja

Period of the project

August 2026 to September 2026



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Declaration

The project titled **“End-to-End Supply Chain and Operational Performance Analysis Across Petroleum, Chemical, and Transportation Sectors Using Data Analytics”** has been mentored by V Keerthana, organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. B.Sreeja

V Keerthana
Data Analyst—EXL

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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Executive Summary

- This project analyzes trade, supply chain, and energy datasets to uncover economic and operational insights for India through an end-to-end analytics pipeline.
- The workflow covers data quality auditing, cleaning, exploratory data analysis, statistical validation, feature engineering, machine learning, SQL querying, and Power BI dashboarding.
- Key findings include a persistent trade deficit that has widened since 2020, logistics delays driven primarily by traffic conditions, and steady but slow growth in renewable energy adoption.
- A Linear Regression model was built to predict trade balance, and three interactive Power BI dashboards (Trade, Supply Chain, Energy) were developed to support ongoing decision-making.
- Recommendations include reducing import dependency, improving logistics efficiency, optimizing operational costs, and increasing investment in renewable energy.



Introduction

Background & Context

In the modern data-driven economy, organizations rely heavily on data analytics to make strategic decisions. This project analyzes multiple datasets related to trade, supply chain, and energy to extract meaningful insights and support business decision-making, with the integration of these three domains providing a comprehensive understanding of economic and operational dynamics, particularly focusing on India.

Problem Statement

Organizations and policymakers often lack an integrated view of how trade performance, supply chain risk, and energy trends interact. This project addresses that gap by building a single analytical pipeline that quantifies trade imbalance, logistics inefficiency, and energy consumption patterns.

Scope & Limitations

- Scope: import/export trade data, supply chain and logistics operations, and energy consumption — primarily India-focused, spanning multiple years of historical data
- Limitation: analysis is based on historical datasets rather than real-time feeds
- Limitation: the predictive model (Linear Regression) is intentionally simple, prioritizing interpretability over maximum predictive accuracy

Innovation Component

The project's innovation lies in integrating three traditionally siloed domains — trade, supply chain, and energy — into a single standardized analytical pipeline, and in translating statistical findings into engineered features, a predictive model, and business-ready Power BI dashboards rather than stopping at descriptive analysis.



Project Objectives

Objectives

- Analyze import-export trends and evaluate trade balance
- Identify inefficiencies and risks in supply chain operations
- Study energy consumption patterns and renewable energy adoption
- Validate insights using statistical methods
- Enhance datasets using feature engineering
- Build a predictive model using machine learning
- Derive business insights using SQL queries
- Present findings through interactive dashboards

Expected Outcomes & Deliverables

- Cleaned and standardized datasets across trade, supply chain, and energy domains
- Exploratory data analysis report with visualized trends
- Statistically validated insights (hypothesis testing results)
- Engineered features for trade, supply chain, and energy datasets
- Trained Linear Regression model predicting trade balance
- SQL query set answering key business questions
- Three Power BI dashboards: Trade Performance, Supply Chain & Logistics, Energy & Sustainability
- Documented business recommendations



Methodology and Results

Methods / Technology Used

- Data quality audit and cleaning (missing values, duplicates, inconsistent formats, aggregated rows)
- Data transformation and standardization (column names, units, categorical values, date/year formats)
- Exploratory data analysis (EDA) across trade, supply chain, and energy
- Statistical analysis: descriptive statistics, correlation analysis, hypothesis testing
- Feature engineering (e.g. Trade Balance, Import-Export Ratio, Cost per Kilometer, Renewable Ratio)
- Machine learning: Linear Regression to predict trade balance
- SQL-based business querying
- Dashboard visualization using Power BI

Tools / Software Used

- Python (pandas, numpy)
- matplotlib / seaborn for visualization
- scipy / statsmodels for statistical testing
- scikit-learn for machine learning
- SQL
- Power BI

Data Collection Approach

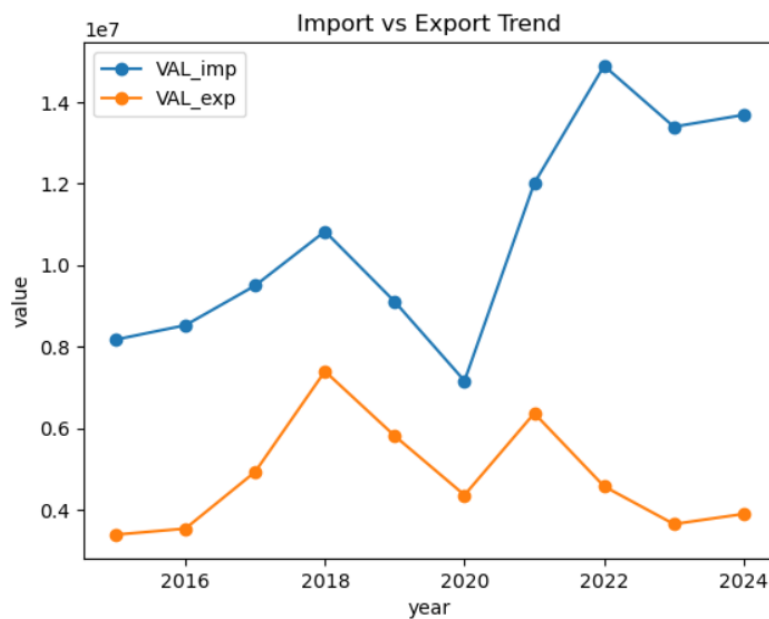
The project uses seven structured datasets — Supply Chain, Petrol, Chemical, Imports, Exports, Logistics, and Energy — containing numerical, categorical, and time-based variables. These were profiled, cleaned, standardized, and merged into unified trade, supply chain, and energy views for analysis.

Project Architecture

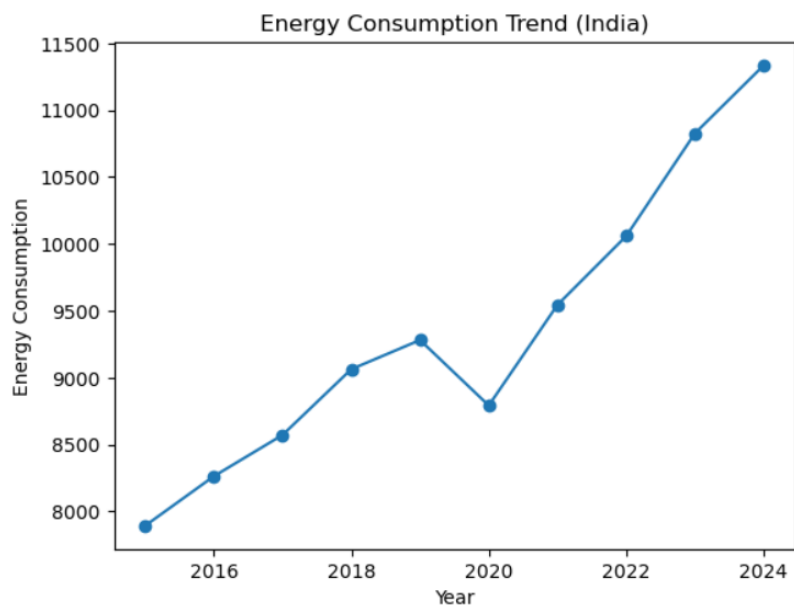
Raw Datasets → Data Quality Audit → Cleaning & Transformation → Standardization → Exploratory Data Analysis → Statistical Validation → Feature Engineering → Machine Learning Model → SQL Business Queries → Power BI Dashboards → Business Recommendations.



Final Project Working Screenshots



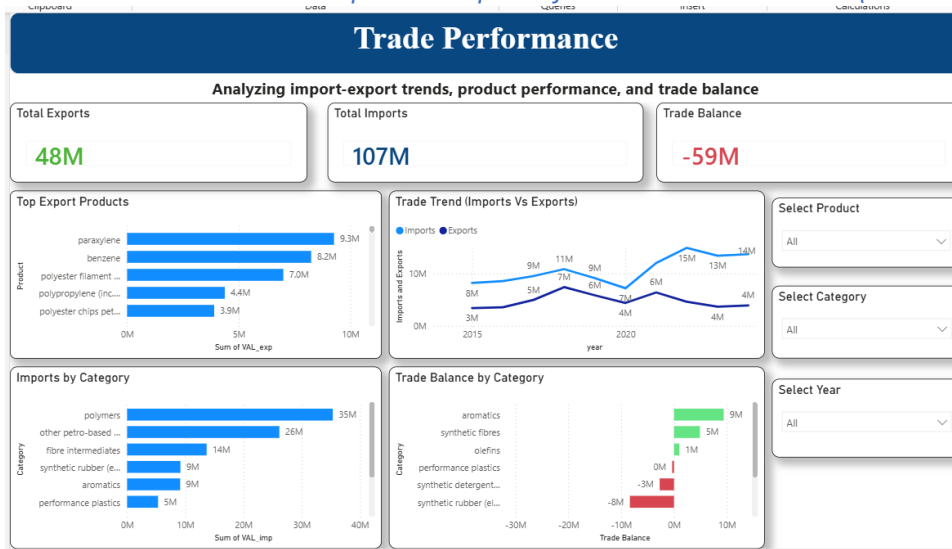
Import vs. Export trend, 2015–2024 — imports consistently exceed exports.



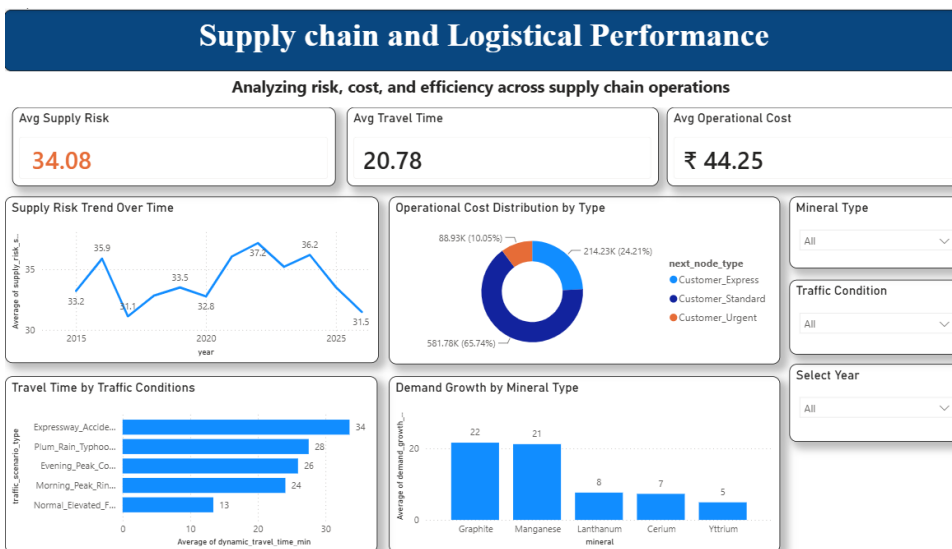
India energy consumption trend, showing steady year-on-year growth.



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Power BI — Trade Performance dashboard.



Power BI — Supply Chain & Logistics dashboard.



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India Energy Consumption & Sustainability Analysis

Tracking consumption trends, renewable growth, and energy mix

Total Energy Consumption

9.36K

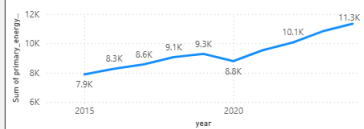
Avg Renewable Energy Share

7.9913

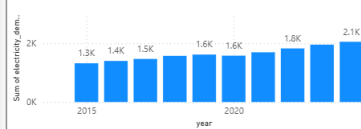
Total Electricity Demand

16.50K

Energy Consumption Trend



Electricity Demand by year



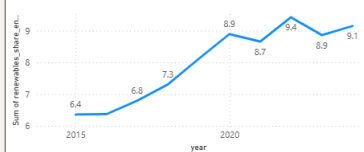
Energy Source Filter

All

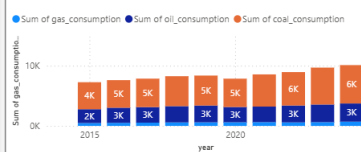
Select Year

All

Renewable Energy Share



Energy Sources Breakdown



Power BI — Energy & Sustainability dashboard.

Project GitHub Link

<https://github.com/sure-trust/BRUNGI-SREEJA-g2-data-analytics.git>



Learning and Reflection

New Learnings

During my internship, I gained hands-on experience in data analysis — building dashboards in Power BI and Tableau, analyzing datasets, and performing hypothesis testing to drive data-informed decisions. This internship gave me practical, hands-on exposure to the core responsibilities of a data analyst

Overall Experience

Overall, it was a great learning experience. I got to work hands-on with real datasets instead of just theoretical problems, which helped me understand how messy and inconsistent real-world data can be. I built dashboards in Power BI and Tableau to present insights clearly to stakeholders, and I also applied statistical methods like hypothesis testing to validate findings before making recommendations. It taught me not just the technical tools, but also how to think critically about data and communicate insights effectively.



Conclusion and Future Scope

Recap of Objectives & Achievements

This project set out to analyze trade, supply chain, and energy data to support data-driven decision-making, and delivered on that goal: a clean, standardized multi-domain dataset; validated statistical insights; an interpretable predictive model for trade balance; a set of business-focused SQL queries; and three Power BI dashboards. Together these confirm a persistent trade imbalance, traffic-driven logistics inefficiency, and gradual renewable energy growth — and translate those findings into concrete business recommendations.

Future Scope

- Use advanced machine learning models to improve predictive accuracy
- Include real-time data feeds for up-to-date monitoring
- Expand the analysis beyond India to a global comparative view
- Improve predictive accuracy through additional features and model tuning